

Introduction to Linear Algebra Chapter 3

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1 Problems

3.2 A. $R = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. $s_1 = \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ and $s_2 = \begin{pmatrix} -2 \\ 0 \\ -6 \\ 1 \end{pmatrix}$. $Rs_1 = Rs_2 = 0$.

3.2 B. A_1 has special solutions e_1, e_2, e_3, e_4 . $R_2 = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$. $s_2 = (-2 \ 1)$.
 r pivots, $n - r$ free, $n - r$ special solutions.

P 1. (a) $A = \begin{pmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix}$. $R_A = \begin{pmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$. x_2, x_4, x_5 are free. x_1, x_3 are pivot.

(b) $B = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 4 & 4 \\ 0 & 8 & 8 \end{pmatrix}$. $R_B = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. x_3 free, x_1, x_2 pivot.

2. For A, $s_2 = (-2, 1, 0, 0, 0)$, $s_4 = (0, 0, -2, 1, 0)$, $s_5 = (0, 0, -3, 0, 1)$. For B, $s_3 = (1, -1, 1)$. 3. $N(A) = c_2s_2 + c_4s_4 + c_5s_5$. $N(B) = c_3s_3$. The $N(M) = Z$ iff M has no special solutions.

4. True.

5. $U_A = \begin{pmatrix} -1 & 3 & 5 \\ 0 & 0 & 0 \end{pmatrix}$. $U_B = \begin{pmatrix} -1 & 3 & 5 \\ 0 & 0 & -3 \end{pmatrix}$. $B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} U_B$.

6. For A, $s_2 = (3, 1, 0)$, $s_3 = (5, 0, 1)$. For B, $s_2 = (3, 1, 0)$. n .

8. $R_A = \begin{pmatrix} 1 & -3 & -5 \\ 0 & 0 & 0 \end{pmatrix}$. $R_B = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. For R_A there's just (1). For R_B , it's I_2 .

9. (a) False, take $A = 0$, all vars are free.

(b) True. If M has a free variable, there's a special solution, so there exists

nonzero vector in $N(M)$, contradiction. $N(M) = Z$.

(c) True. num pivot + num free = n , so num pivot is at most n .

(d) True. There is at most one pivot per row, and there are m rows.

14. x_5 . $s_5 = (-1, 0, 0, 0, 1)$.

16. 5 pivots. No pivots (use otherwise). M is invertible, so M has 5 pivots.

20. Column 5 has no pivot and is therefore free. C_5 is a combination of earlier columns. $s_5 = (1, 0, 1, 0, 1)$. $N(M)$ is multiples of s_5 .

24. Not possible. $C(M)$ and $N(M)$ are both subspaces of $\mathbb{R}^3$. $C(M)$ has two linearly independent vectors, and same with the nullspace. Then $\mathbb{R}^3$ has a basis with four elements, contradiction.

26. $A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$. Null space is $c(1, 1)$. Column space is $c(1, 1)$.

27. Then there would be an equal number of pivots and free variables. The sum of the number of pivots and free variables is odd, however.

30. (a) Take $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. $s_2 = (-1, 1, 0)$, $s_3 = (-1, 0, 1)$. $N = c_2 s_2 + c_3 s_3$.

$A^T = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. Reduced becomes only $a_{11} = 1$. $s_2 = (0, 1, 0)$, $s_3 = (0, 0, 1)$.
 $N = c_2 s_2 + c_3 s_3$.

Chapter 3.3

$A = \begin{pmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $r = 2$.

$R = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. $r = 3$, $s_4 = (1, 1, 1, 1)$.

3.3 B. $A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$. $ad = bc$, $d = bc/a$. $A = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \begin{pmatrix} 1 & b/a \end{pmatrix}$.

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 2 & 3 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}. \text{ We have } A = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 \end{pmatrix} \text{ in the desired form.}$$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & c \end{pmatrix}. \quad R = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \text{ if } c \neq 4. \quad \text{Else } R = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \text{ If}$$

$$c \neq 4, r = 2. \quad s_2 = (-2, 1, 0) \text{ and } N = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}. \text{ If } c = 4, r = 1. \quad s_2 = (-2, 1, 0)$$

$$\text{and } s_3 = (-1, 0, 1), \text{ for which } N = \begin{pmatrix} -2 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad A = \begin{pmatrix} c & c \\ c & c \end{pmatrix}. \text{ If } c = 0, r = 0,$$

$$s_1 = (1, 0), s_2 = (0, 1). \quad N = I_2. \text{ If } c \neq 0, R = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}. \quad r = 1, s_2 = (-1, 1),$$

$$N = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

P 1. (a) yes (b) no (c) yes (d) no

$$\text{P 3. } R = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad R = \begin{pmatrix} 1 & 2 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad R = \begin{pmatrix} 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\text{P 7. } s_3 = (-2, -4, 1, 0), s_4 = (-3, -5, 0, 1). \quad s_2 = (0, 1, 0), s_3 = (0, 0, 1).$$

P 13. When P is reduced to echelon form U' , there are still r pivots, so P is rank r .

$$\text{P 8. } A = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 4 & 8 & 16 \end{pmatrix}. \quad B = \begin{pmatrix} 3 & 9 & -9/2 \\ 1 & 3 & -3/2 \\ 2 & 6 & -3 \end{pmatrix} \quad M = \begin{pmatrix} a & b \\ c & bc/a \end{pmatrix}.$$

$$\text{P 16. } (uv^\top)(wz^\top) = (v^\top w)(uz^\top). \quad uz^\top = 0 \text{ or } v^\top w = 0.$$

P 17. $\text{Rank}(AB) \leq \text{Rank}(A)$. Let $c_1, c_2, \dots, c_n$ be the columns of B . Suppose $c_j = x_1 c_{a_1} + x_2 c_{a_2} + \dots + x_k c_{a_k}$, where each $a_l < j$. Then $Ac_j = x_1(Ac_{a_1}) + \dots + x_k(Ac_{a_k})$, so the j th column of AB is also free. AB and B have the same number of columns, so AB has at most as many pivots as B .

P 19. $n = \text{Rank}(I) = \text{Rank}(AB) \leq \text{Rank}(A)$. Since $\text{Rank}(A) \leq n$ as A has n columns, $\text{Rank}(A) = n$. Then A is invertible, so B is its two-sided inverse.

P 21. Suppose A and B have the same reduced row echelon form. (a) A is off from B by an invertible matrix, so their row spaces are the same. Null space is dependent on special solutions, which is dependent on RREF. Since RREF is the same, nullspace is the same.

P 22. $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 8 \end{pmatrix}.$

$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1/4 & 1/4 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 1 & 8 & 1 \end{pmatrix}.$$

Now, $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 4 & 0 \\ 1 & 8 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \\ 8 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}. B = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$

$$R = \begin{pmatrix} I & F \\ 0 & 0 \end{pmatrix}.$$

(a) I is $r \times r$, F is $r \times (n - r)$, 0_{left} is $(m - r) \times r$, 0_{right} is $(m - r) \times (n - r)$.

(b) If $r = m$, $R = \begin{pmatrix} I & F \end{pmatrix}$. $B = \begin{pmatrix} I \\ 0 \end{pmatrix}$. $RB = I^2 + F0 = I$. If $r = n$, $R = \begin{pmatrix} I \\ 0 \end{pmatrix}$.

$$\begin{pmatrix} I & 0 \end{pmatrix} \begin{pmatrix} I \\ 0 \end{pmatrix} = I.$$

$R^T = \begin{pmatrix} I & 0 \\ F^T & 0 \end{pmatrix}$. I is $r \times r$. F^T is $(n - r) \times r$. 0_{top} is $r \times (m - r)$. 0_{bottom} is $(n - r) \times (m - r)$. The RREF becomes I in top left, 0s elsewhere

$$R^T R = \begin{pmatrix} I & F \\ F^T & F^T F \end{pmatrix}. \text{ RREF is } R, N \text{ is same for both.}$$

P 28. Identity in top left, zeroes elsewhere.

$$B = \begin{pmatrix} 1 & 1 & b_1 \\ 1 & 2 & b_2 \\ -2 & -3 & b_3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & b_1 \\ 0 & 1 & b_2 - b_1 \\ 0 & -1 & 2b_1 + b_3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & b_1 \\ 0 & 1 & b_2 - b_1 \\ 0 & 0 & b_1 + b_2 + b_3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 2b_1 - b_2 \\ 0 & 1 & b_2 - b_1 \\ 0 & 0 & b_1 + b_2 + b_3 \end{pmatrix}.$$

Require $b_1 + b_2 + b_3 = 0$. $x = \begin{pmatrix} 2b_1 - b_2 \\ -b_1 + b_2 \end{pmatrix}$. $M = \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 & 3 \\ 0 & 1 & -2 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & 1 \end{pmatrix}$. $m = r = 2$. $x_p = (2, 1, 0)$. $x_n = c(-3, 2, 1)$.

$$\begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 2 & 4 & 8 & 12 & b_2 \\ 3 & 6 & 7 & 13 & b_3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & -2b_1 + b_2 \\ 0 & 0 & -2 & -2 & -3b_1 + b_3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 2 & 2 & -2b_1 + b_2 \\ 0 & 0 & 0 & 0 & -5b_1 + b_2 + b_3 \end{pmatrix}.$$

$$2. -5b_1 + b_2 + b_3 = 0.$$

3. Plane in $\mathbb{R}^3$. $C(A) = x_1c_1 + x_3c_3$.

4. $N = x_2(-2, 1, 0, 0) + x_4(-2, 0, -1, 1)$. $-9(1, 2, 3) + 3(3, 8, 7) = (0, 6, -6)$.
Complete: $(-9, 0, 3, 0) + x_2s_2 + x_4s_4$.

$$R = \begin{pmatrix} 1 & 2 & 3 & 5 & b_1 \\ 0 & 0 & 1 & 1 & -b_1 + b_2/2 \\ 0 & 0 & 0 & 0 & -5b_1 + b_2 + b_3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 & 2 & 4b_1 - 3b_2/2 \\ 0 & 0 & 1 & 1 & -b_1 + b_2/2 \\ 0 & 0 & 0 & 0 & -5b_1 + b_2 + b_3 \end{pmatrix}. x_p = (4b_1 - 3b_2/2, 0, -b_1 + b_2/2, 0). x_2 = (4b_1 - 3b_2/2 - 2, 1, -b_1 + b_2/2, 0). x_4 = (4b_1 - 3b_2/2, 0, -b_1 + b_2/2 - 1, 1).$$

1. A has full-column rank, b is in the column space of A .

2. The nullspace of A is all multiples of $(1, 1)$. $n = 2$ and column 2 is the negative of c_1 .

3. Not much info.

4. $n = 3$, $c_3 = -c_1$. The nullspace of A is all multiples of $(1, 0, 1)$.

5. $n \geq m$.

3.4 C.

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 & 4 \\ 2 & 4 & 4 & 8 & 2 \\ 4 & 8 & 6 & 8 & 10 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 2 & 8 & -6 \\ 0 & 0 & 2 & 8 & -6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 & -4 & 7 \\ 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$x_p = (7, 0, -3, 0)$. $x_2 = (-2, 1, 0, 0)$. $x_4 = (4, 0, -4, 1)$. $x = x_p + c_2x_2 + c_4x_4$. $y_2 - 2y_1 = y_3 - 4y_1$, so $2y_1 + y_2 - y_3 = 0$. Check $2(4) + 2 - 10 = 0$. This ensures that the last row in row reduction becomes $0 = 0$. We find some y such that y is perpendicular to each column spanning $C(A)$. Then y is perpendicular to all of $C(A)$. For $b = Ax$ to be solvable, b lies in $C(A)$. b must also be perpendicular to y .

P 2.

$$A = \begin{pmatrix} 2 & 1 & 3 & 10 \\ 6 & 3 & 9 & 30 \\ 4 & 2 & 6 & 20 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 3 & 10 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1/2 & 3/2 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$x_p = (5, 0, 0)$. $x_2 = (-1/2, 1, 0)$, $x_3 = (-3/2, 0, 1)$. $x = x_p + c_2x_2 + c_3x_3$.
Require $b_2 = 3b_1$ and $b_3 = 2b_1$. These conditions put b into $C(A)$.

P 4.

$$A = \begin{pmatrix} 1 & 3 & 1 & 2 & 1 \\ 2 & 6 & 4 & 8 & 3 \\ 0 & 0 & 2 & 4 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 1 & 2 & 1 \\ 0 & 0 & 2 & 4 & 1 \\ 0 & 0 & 2 & 4 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$x_p = (1/2, 0, 1/2, 0)$. $x_2 = (-3, 1, 0, 0)$, $x_4 = (0, 0, -2, 1)$. $x = x_p + c_2x_2 + c_4x_4$.

P 6.

$$A = \begin{pmatrix} 1 & 2 & b_1 \\ 2 & 4 & b_2 \\ 2 & 5 & b_3 \\ 3 & 9 & b_4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & b_1 \\ 0 & 0 & -2b_1 + b_2 \\ 0 & 1 & -2b_1 + b_3 \\ 0 & 3 & -3b_1 + b_4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 5b_1 - 2b_3 \\ 0 & 0 & -2b_1 + b_2 \\ 0 & 1 & -2b_1 + b_3 \\ 0 & 0 & 3b_1 - 3b_3 + b_4 \end{pmatrix}.$$

Require $-2b_1 + b_2 = 0$, $3b_1 - 3b_3 + b_4 = 0$. Unique solution: $x = (5b_1 - 2b_3, -2b_1 + b_3)$.

P 36. To speak about same b , same solutions x , A, C have the same dimensions. Pick b to be the first column of A . Then $x = (1, 0, \dots, 0)$ is a solution of both, so the first columns match. In general, all columns match, so $A = C$.

It is the dimension of the column space that equals the rank of the matrix.